How layer imputation works

The workflow generates 100 plausible, ordered sets of contact elevations at the intersection locations. These completed datasets then condition surface and thickness simulations across the study area.

1 Prepare and rotate the geology

Notebook 01 limits the study to Phase 3 and prepares buffered meshes and composites. Notebook 02 fits a plane to Chert 3 and rotates coordinates to (u, v, w): u and v lie along the plane; w is perpendicular to it. This rotates the geology but does not fully unfold curved layers.

2 Build a sparse contact table

Drillhole interval segments are intersected with existing stratigraphic meshes. Each retained top intersection becomes a row with one known contact elevation and nine missing elevations. Rows represent individual intersections, not whole drillholes. The known contacts are mesh-derived, rather than newly picked from lithology logs.

3 Complete surfaces from the middle outward

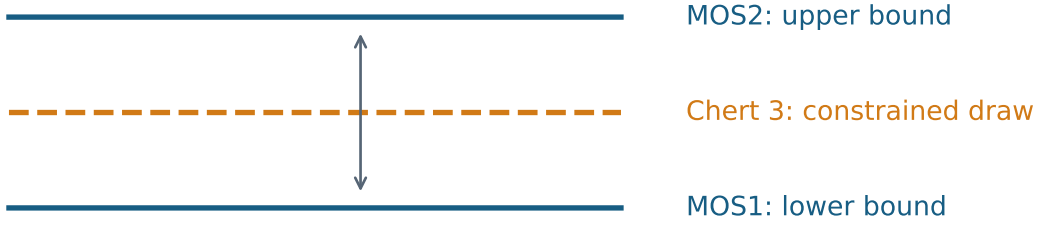
The imputation order is Chert 3, MOS1, MOS2, Chert 4, Chert 2, LOS, UOS, Chert 1, Chert 5, then UBB. Previously imputed surfaces constrain later surfaces separately within each realization.

4 Bound each missing contact

At a fixed (u, v), the nearest available surface below and above the target supplies its lower and upper limits. At a known target contact, both limits equal the known elevation. Without a bound on one side, the code uses the target contact minimum or maximum, extended by 0.001.

5 Draw spatially conditioned elevations

For each surface, the code declusters known elevations, smooths their distribution with KDE, transforms to normal scores, and fits a directional variogram. ConstrainedImputer draws values within the transformed bounds; inverse transformation returns elevations in original units. Kriging cross-validation is a diagnostic, not the imputation itself.



Schematic at one (u, v): lower bound <= target elevation <= upper bound.

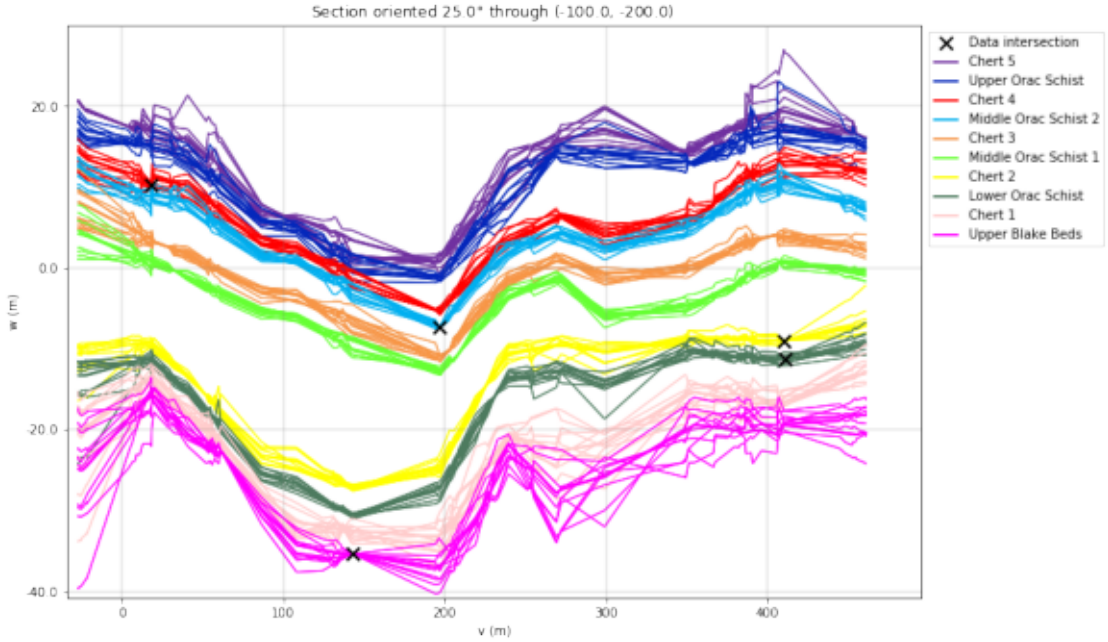
From contacts to simulated layers

6 Calculate adjacent thicknesses

After completing ten surface elevations at every location, calculate nine thicknesses as upper w minus lower w. Each realization supplies Chert 3 elevation plus nine thicknesses. These are separations along the rotated w direction, not vertical thicknesses in original coordinates.

7 Simulate and reconstruct the stack

Notebook 03 uses each completion to condition 2D spatial simulation, with distribution transforms, trend removal, spatial bootstrap, and a multivariate transform. After simulation, smoothing with conditioning correction, and inverse transforms, it reconstructs surfaces by adding thicknesses above Chert 3 and subtracting them below. The surfaces classify the 3D grid and produce layer probabilities and tonnage variability.



Saved output from Notebook 02, section-check cell: black crosses are original mesh-derived intersections; colored traces show alternative completed surfaces. Lines connect completed points through triangulation. This historical plot was extracted without rerunning the notebook.

Interpretation and limits

The method assumes an ordered stack with one elevation per surface at each (u, v). Overturned folds or repeated faulted sequences challenge that assumption. Input mesh interpretation and the sequential imputation order influence the result.

Notebook 03 contains an outdated sentence describing UBB as the base. Its executable reconstruction uses Chert 3. Saved outputs record RMSP 1.8.14; available API documentation is for 1.18.3, so historical internal defaults were not verified.

Sources reviewed

06-StratigraphicSimulation: 01-ModelAndCompositeExtent.ipynb; 02-IntersectionImputation.ipynb; 03-SurfaceThicknessSimulation.ipynb. Tools: kde_fit.py, utilities.py, project_parameters.py and report_params.json.